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CLIMATE & SUSTAINABILITY

Stock and Flow — What a Tonne of Methane Is Worth

Every climate inventory on earth converts methane into carbon dioxide using a single number, and that number was published as an illustration of why no such number exists.

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Somewhere in a spreadsheet maintained by every government that reports to the United Nations, a quantity of methane is being multiplied by twenty-seven. The product is entered as carbon dioxide equivalent, added to a national total, and compared against a target. The number twenty-seven does an extraordinary amount of work in the world. It decides whether a country is on track, whether a dairy sector is a problem or a rounding error, and whether a gas leak in Turkmenistan matters more or less than a coal plant in Poland.

It is worth knowing where it comes from. The Global Warming Potential was introduced in 1990, in the first assessment report of the Intergovernmental Panel on Climate Change, and the report was candid about what it was doing. There was, it said, no universally accepted method for combining all the relevant factors into a single index; a simple approach had been adopted to illustrate the difficulties inherent in the concept [1]. Three time horizons were offered — twenty years, one hundred, five hundred — precisely because no one of them was right. The Kyoto Protocol picked the middle one in 1997 without explaining why, and the Paris Agreement's transparency rules made it mandatory in 2018, requiring every party to report its inventory using hundred-year Global Warming Potentials from the IPCC's assessment reports [2]. An illustration of a difficulty had become a legal obligation.

The difficulty did not go away. It is now the most consequential unresolved argument in climate accounting, and it turns entirely on the fact that methane leaves.

A Gas That Leaves

Carbon dioxide, for practical purposes, accumulates. A fraction of any pulse emitted today is still in the atmosphere in a thousand years, which is why the physics licenses talking about a carbon budget at all: peak warming depends on the cumulative total, not on the rate. Methane behaves nothing like this. Hydroxyl radicals in the troposphere destroy it, and the destruction is proportional to how much is there, which makes the atmosphere a leaky bucket rather than a reservoir.

Write B for the mass of methane in the air above the pre-industrial background, E for the emission rate, and τ for the time constant of the loss:

$$\frac{dB}{dt} = E(t) - \frac{B}{\tau}$$

The consequence is immediate. Hold E constant and B does not grow without bound; it converges on $B = E\tau$ and stops. A constant methane source builds a stock of methane in the atmosphere once, over about three decades, and thereafter simply maintains it. This is not an approximation of convenience. Take the observed global mean in 2024, 1,921.79 parts per billion, convert at the standard 2.75 teragrams of methane per part per billion, and the atmosphere holds about 5,285 teragrams [3]. Divide by the Global Carbon Project's estimate of total emissions for the 2010s, 575 teragrams a year, and you recover a residence time of 9.2 years [4]. The assessed atmospheric lifetime is 9.1 years. The bucket really is leaking at the rate the model says.

The 11.8-year figure that appears in the metric calculations is a slightly different quantity, and the gap between them is a piece of chemistry worth pausing on. Methane consumes the hydroxyl radicals that destroy it, so adding methane to the atmosphere lengthens the lifetime of the methane already there. A perturbation therefore decays more slowly than the bulk stock turns over — 11.8 years rather than 9.1 [3]. The gas is mildly self-protecting.

None of this makes methane a minor gas. It is responsible for roughly 0.5 degrees of the warming observed over 2010–2019 relative to 1850–1900, against about 0.8 degrees from carbon dioxide, and concentrations are now 2.6 times the pre-industrial 729 parts per billion [3]. In 2021 the global mean concentration rose by 17 parts per billion, the largest annual increase since systematic global measurement began in 1983 [5]. The problem is not that methane is unimportant. The problem is that a stock pollutant and a flow pollutant are being added together, and addition requires a common unit.

What the Integral Is Doing

The Global Warming Potential supplies that unit by integrating. For a gas x , the absolute potential over a horizon H is the radiative forcing caused by one kilogram, accumulated:

$$\text{AGWP}_x(H) = \int_0^H A_x f_x(t) dt$$

where A_x is the forcing per kilogram and $f_x(t)$ the fraction still airborne after t years. The Global Warming Potential is then the ratio of that integral to the same integral for carbon dioxide. Everything contentious about the metric lives in the upper limit.

For methane, $f(t) = e^{-t/\tau}$, so the integral saturates: it reaches 9.6 year-units by twenty years and 11.8 by a hundred, which is to say that four-fifths of methane's entire lifetime contribution is delivered inside the first two decades and almost nothing arrives after that. For carbon dioxide the airborne fraction is a sum of four terms — one of which never decays at all — and the integral keeps climbing, from 13.9 year-units at twenty years to 52.0 at a hundred [6]. The denominator grows almost fourfold while the numerator barely moves.

That single fact predicts the whole structure of the published tables. Taking the ratio of ratios gives a factor of 3.06 between the twenty-year and hundred-year potentials for methane; the assessed values, roughly 80 and 27 for methane of biological origin, give 3.0 [3]. A two-line calculation reproduces the official number to about four per cent, and the residue is chemistry the simple model leaves out — the ozone that methane makes, the stratospheric water vapour it becomes, and, for methane of fossil origin, the carbon dioxide it eventually oxidises into, which is why fossil methane is assessed at 29.8 rather than 27.

WHY SHOULD THE CHOICE OF HORIZON BE A SCIENTIFIC QUESTION AT ALL?

It is not one, and that is the uncomfortable part. Extending the horizon does not make the estimate more accurate; it changes the question being asked, from *how much heating over the next twenty years* to *how much over the next century*. Keith Shine, who has spent a career on these metrics, argued in 2009 that the Global Warming Potential had never received the interdisciplinary trial it needed, and that the choice of horizon smuggles in a value judgement about how much the near term matters relative to the far one — a discount rate wearing the clothes of a physical constant [7]. The hundred-year horizon is not a measurement. It is a preference, adopted by a treaty, and then reported to four significant figures.

The Star

In 2018 a group at Oxford proposed a different use of the same numbers. Their observation was simple: if a change in the *rate* of a short-lived gas behaves like a one-off *pulse* of a long-lived one, then the right thing to multiply is not the emission but its recent change [8]. The metric that emerged, refined in 2019 and again in 2021, is written GWP* and reads:

$$E^*(t) = \text{GWP}_{100} \left[r \frac{E(t) - E(t - \Delta t)}{\Delta t} H + s E(t) \right]$$

with $H = 100$ years, $\Delta t = 20$ years, and weights $r = 0.75$ and $s = 0.25$ found to track the modelled warming from methane across a wide range of trajectories [9], [10]. Applied to real inventories it behaves as advertised: conventional hundred-year accounting misreports the warming from methane in most scenarios, and badly whenever emissions are flat or falling [11]. The first bracketed term is a flow term, responding to how fast emissions are changing; the second is a stock term, acknowledging that even an unchanging forcing goes on warming the ocean for a long time. Since $rH/\Delta t = 3.75$ exactly, the arithmetic is unusually clean.

Set E^* to zero and ask what emissions path earns a charge of nothing. The condition is that emissions today stand in the ratio **3.75 : 4** to emissions twenty years ago — that is, exactly fifteen-sixteenths. A methane source that shrinks by 6.25 per cent over any twenty-year window, about 0.32 per cent a year, is

adding no further warming, and the metric says so by returning zero. Cut faster than that and the number goes negative: the source is a cooling agent, not because methane cools anything, but because its declining stock is releasing heating the world had already been committed to.

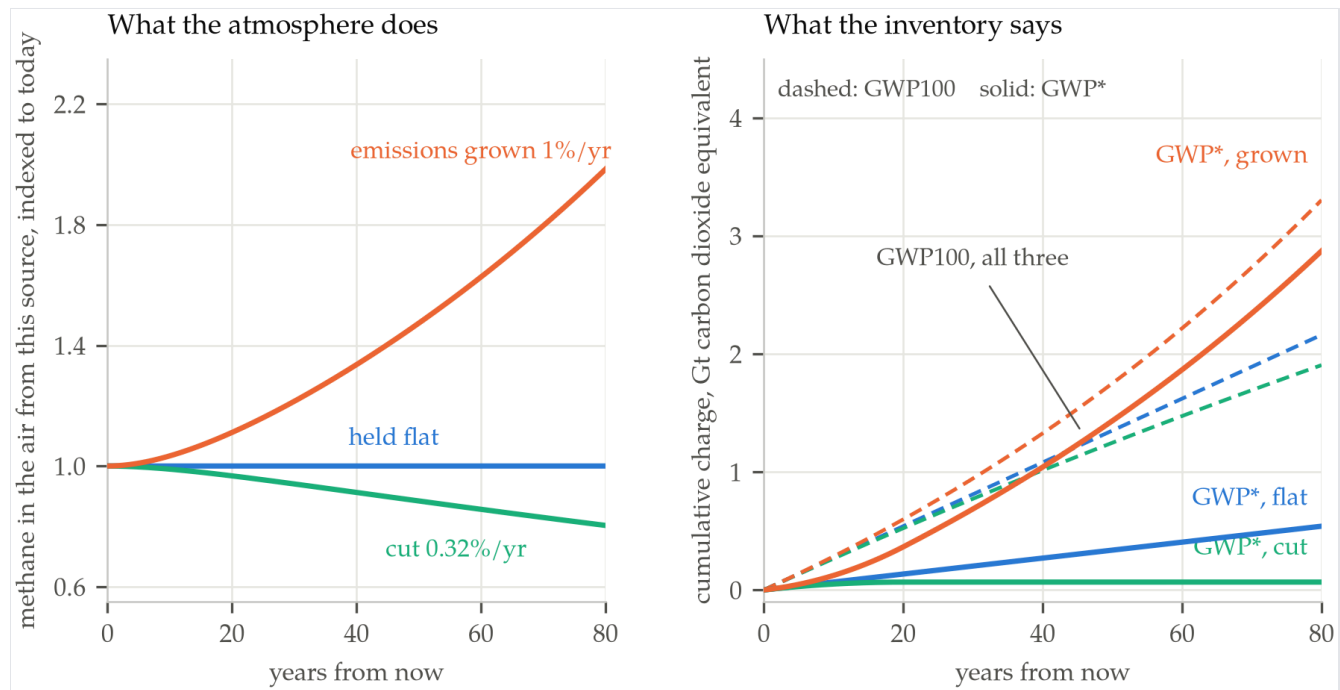


FIGURE Left: a source emitting one million tonnes of methane a year, indexed to its present contribution, under three futures. Right: what the same three futures are charged, cumulatively, under the hundred-year Global Warming Potential (dashed) and under GWP* (solid). The shrinking source is still billed nearly two billion tonnes.

The other end of the scale is more startling. A brand-new methane source — a herd that did not exist, a field newly drilled — has $E(t - \Delta t) = 0$, so its flow term is at maximum and it is charged $3.75 + 0.25 = 4$ times the conventional figure for its first two decades. An identical source that has been running for fifty years is charged $s = 0.25$ times the conventional figure. Same gas, same tonnage, same molecules arriving in the same atmosphere, and a sixteen-to-one difference in the bill.

A source emitting one Mt of methane a year	GWP100	GWP*
Steady for decades	27	6.8
Newly established, first 20 years	27	108
Falling 0.32 per cent a year	27	0
Falling 3 per cent a year	27	–78
Rising 1 per cent a year	27	25

Charges in millions of tonnes of carbon dioxide equivalent a year. The constancy of the middle column is the entire argument: five physically different situations, one answer.

It would be a mistake to read any of this as an argument for leaving methane alone. The 0.32 per cent a year that zeroes GWP* is the rate at which a source stops making things worse, not the rate at which anyone should be satisfied, and it is roughly a tenth of what is currently on the table. The Global Methane Pledge, launched at Glasgow in 2021 and now carrying well over a hundred signatories, asks for 30 per cent below 2020 levels by 2030 — a decline of 3.5 per cent a year. The assessment that prompted it found that measures already available could cut human methane emissions by about 45 per cent this decade and avoid nearly 0.3 degrees of warming by the 2040s [18]. Because methane's stock responds within decades rather than centuries, cutting it is the only lever that changes temperature on a timescale a government can be held to. GWP* does not weaken that case; if anything it sharpens it, because it is the metric that actually rewards a cut for arriving early.

Whose Baseline?

The sixteen-to-one ratio is not a bug that better calibration would remove. It is what warming-equivalence means, and it is also, unavoidably, a distributional claim. In 2019 Joeri Rogelj and Carl-Friedrich Schleussner pointed out that applying GWP* below the global level rewards whoever was already emitting: a country with a large, stable methane sector is credited for sustaining warming it caused earlier, while a country whose herds are only now growing is charged four times over for reaching the same level [12]. Historical emissions become an entitlement. The technical term for this in every other corner of climate policy is grandfathering, and it is normally the thing being argued against.

The reply, from Michelle Cain, Keith Shine and colleagues in 2021, did not deny the arithmetic. It denied that the arithmetic was the place to fix the problem: a metric that reports warming accurately should report it accurately, and it is not for scientists to encode their own view of other people's interests in a conversion factor [13]. Equity, on this account, belongs in the target — in who is required to reduce by how much — and not in the unit. It is a genuinely hard disagreement, because both halves are right about their own half.

Meanwhile the practical worry is less philosophical. Because GWP* measures change from a baseline, the choice of baseline year decides the answer, and the same herd can be described as warming, neutral or cooling depending on where the clock starts. A 2025 assessment in *Environmental Research Letters* proposed an alternative — a 44-year lagged cumulative sum of methane emissions, which tracks temperature about as well as GWP* across scenarios, cannot be gamed into a negative number while emissions are positive, and sits more comfortably with the way the Paris Agreement is written [14]. That such a proposal exists at all is a reminder that GWP* is not the end of this argument, only its most successful move so far.

IF THE HORIZONS ARE ARBITRARY AND THE BASELINES ARE CONTESTED, WHY AGGREGATE THE GASES AT ALL?

The honest answer is that nobody has to. The assessment reports themselves note that treating short-lived and long-lived gases as separate pathways, and reporting them separately, represents warming at least as well as any single aggregate does [3]. A country could publish two numbers and let the reader do the work. What stops it is not physics but institutions: a single figure is what an emissions-trading scheme prices, what a headline reports, and what a treaty compares.

What Wellington Decided

One country did split them. New Zealand's Climate Change Response (Zero Carbon) Amendment Act 2019 set net zero by 2050 for the long-lived gases and a separate target for biogenic methane — 10 per cent below 2017 levels by 2030, and 24 to 47 per cent below by 2050 [15]. In a country where methane from ruminants is close to half the inventory, the split was not a technicality; it was the whole design.

Then the logic was applied a second time. An independent scientific review, asked to say what target would be consistent with New Zealand's methane causing no additional warming, reported that in a world moving fast enough to hold 1.5 degrees, a 24 per cent cut by 2050 would do it, and that under slower global scenarios something nearer 14 per cent would suffice [16]. In December 2025 Parliament lowered the statutory 2050 range to 14 to 24 per cent [17]. The upper bound of the new target is the lower bound of the old one.

Whether that is a rigorous application of atmospheric physics or an agricultural lobby's best week in a decade is exactly the question, and the arithmetic does not settle it. What the arithmetic does establish is narrower and more interesting: a 14 per cent cut spread over thirty-three years is a decline of 0.46 per cent a year, comfortably steeper than the 0.32 per cent that zeroes the metric, so the target is not, on its face, an accounting trick. It is also true that "no additional warming" takes as given all the warming already caused, and 2017 is a baseline chosen by legislators rather than by the atmosphere.

Both of those statements are consequences of choosing a unit. That is the durable lesson of the whole affair, and it generalises well beyond methane: whenever incommensurable things are added together — a tonne now against a tonne later, a species against a hectare, a life-year against a pound — the conversion factor is where the ethics go to hide. The 1990 assessment said as much on the page where it introduced the number. Thirty-six years and one binding treaty later, the difficulty it meant to illustrate is still exactly where it left it.

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